Multivariate LSTM-Based Forecasting of Water Demand, Supply, and Storage with Sectoral and Stress Analysis

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***Abstract*—Accurate forecasting of water demand and supply is essential for sustainable water resource management, particularly in regions facing increasing climatic variability and population growth.**

This paper proposes a multivariate Long Short-Term Memory (LSTM) model to jointly forecast water demand, supply, and reservoir storage using time-series data. The model predicts sectoral demand (domestic, agricultural, industrial), total supply, and storage in a unified framework.

The approach captures nonlinear relationships between climatic variables and water consumption patterns while maintaining temporal consistency. In addition, a water stress index is introduced to assess system pressure over time.

Experimental results demonstrate strong predictive performance and good generalization across validation and test periods. The proposed framework provides an interpretable and practical tool for water resource planning and decision-making.

***Index Terms*—Water demand forecasting, LSTM, time-series prediction, hydrological modeling, water stress**

I. Introduction

Water resource management is increasingly challenged by population growth, climate variability, and rising demand across multiple sectors. Accurate forecasting of water demand and supply is essential to ensure sustainable planning, prevent shortages, and support informed decision-making in water-stressed regions.

Traditional forecasting approaches often rely on statistical models that struggle to capture nonlinear relationships and complex temporal dependencies present in hydrological systems. In recent years, deep learning models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in modeling time-series data due to their ability to learn long-term dependencies and dynamic patterns.

In this work, we propose a multivariate LSTM-based framework for jointly forecasting water demand, supply, and reservoir storage. Water demand is decomposed into three sectors—domestic, agricultural, and industrial—to capture heterogeneous consumption behaviors and improve interpretability. Unlike approaches that directly rely on raw storage observations, reservoir storage is first reconstructed using a proxy-based simulation to correct inconsistencies in the original dataset. The LSTM model is then trained to learn this reconstructed storage alongside demand and supply, ensuring coherent modeling of system dynamics.

To further enhance decision-making capabilities, a water stress index is introduced to evaluate the balance between demand and effective supply over time. This indicator allows the identification of critical periods where the system operates under high pressure and supports risk-aware water management strategies.

The main contributions of this work are summarized as follows:

- A multivariate LSTM framework for joint forecasting of demand, supply, and storage in a unified pipeline.
- A sectoral decomposition of water demand to capture heterogeneous consumption patterns.
- A proxy-based reconstruction of reservoir storage to ensure physically consistent learning.
- The integration of a water stress index for assessing system pressure and identifying critical periods.

The remainder of this paper is organized as follows. Section II describes the dataset and preprocessing steps. Section III presents the model architecture. Section IV introduces sectoral demand modeling. Section V analyzes the water stress index. Section VI discusses the results, and Section VII explores policy scenarios. Finally, Section VIII concludes the study and outlines future research directions.

II. Dataset and Preprocessing

The dataset consists of monthly observations from 2000 to 2035 for the Tangier region. It includes climatic, demographic, and hydrological variables such as rainfall, temperature, demand, supply, and storage.

The data is chronologically split into:

- Training: 2000–2021
- Validation: 2022–2025
- Testing: 2026–2035

All variables are normalized using Min-Max scaling. A sliding window of 12 months is used to construct input sequences, allowing the model to capture yearly seasonality.

III. Model Architecture

The model is based on a multivariate LSTM network designed for multi-output time-series forecasting.

The model predicts:

- Domestic demand
- Agricultural demand
- Industrial demand
- Total supply
- Reservoir storage

The architecture includes:

- LSTM layer (32 units)
- Dropout layer (0.2)
- Dense layer (ReLU activation)
- Output layer (5 neurons)

The model is trained using the Huber loss function, which improves robustness to outliers. Early stopping is applied to prevent overfitting.

IV. Sectoral Demand Modeling

Water demand is decomposed into three sectors:

- Domestic
- Agricultural
- Industrial

Total demand is computed as:

$$Demand = Domestic + Agricultural + Industrial \quad (1)$$

Each sector exhibits distinct behavior:

- Domestic demand is stable and population-driven
- Agricultural demand is highly seasonal
- Industrial demand is smoother and less variable

This decomposition improves interpretability and allows more detailed analysis of water consumption.

V. Water Stress Analysis

The water stress index provides a compact and interpretable indicator of the balance between water demand and available supply, reflecting the level of pressure exerted on the system over time.

Figure 1 illustrates both the actual and predicted stress indices over the forecast horizon (2026–2035), together with risk thresholds and highlighted critical periods.

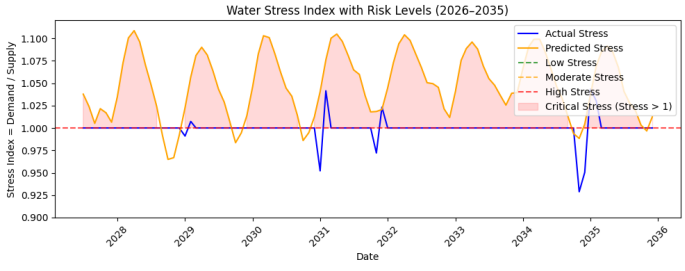


Fig. 1. Water stress index with risk levels and critical periods (2026–2035)

The stress index consistently fluctuates around the critical threshold of 1, indicating that the system operates near its capacity limits throughout the forecast period. This suggests a structurally constrained system where demand and supply remain closely balanced.

Clear seasonal patterns are observed, with recurring peaks corresponding to periods of increased water demand. During these intervals, the predicted stress index frequently exceeds the critical threshold, as highlighted in red in Figure 1. These episodes represent potential risk zones where demand surpasses effective supply, increasing the likelihood of shortages.

The predicted stress curve follows the overall temporal dynamics of the system while remaining smoother than the actual stress. This smoothing effect reflects the model's tendency to capture dominant trends while attenuating short-term fluctuations, which may lead to slight underestimation of extreme stress events.

For predicted values, an adjustment factor is applied to supply to account for distribution losses, evaporation, and operational inefficiencies. The effective supply is therefore modeled as a fraction of the predicted supply, providing a more realistic estimate of system pressure under real-world conditions.

A. Interpretation of Risk Levels

To enhance interpretability, three qualitative stress levels are defined:

- **Low stress** ($Stress < 0.5$)**:** the system operates well below capacity, with sufficient available resources.
- **Moderate stress** ($0.5 \leq Stress < 0.8$)**:** increasing pressure on the system, requiring monitoring and potential mitigation measures.
- **High stress** ($Stress \geq 1$)**:** demand exceeds effective supply, indicating critical conditions and potential shortages.

The results show that the system rarely operates in low-stress conditions and remains predominantly within moderate to high stress regimes. This behavior highlights the structural vulnerability of the system under projected demand patterns, where even small perturbations in demand or supply could lead to critical imbalances.

Overall, the stress analysis emphasizes the importance of proactive water management strategies, including demand regulation, efficiency improvements, and adaptive planning, to ensure system resilience under future conditions.

B. Implications for Water Management

The stress analysis reveals that future water availability is tightly constrained relative to demand. The frequent occurrence of stress values above 1 suggests that even small increases in demand or reductions in supply could lead to critical shortages.

These findings emphasize the importance of proactive water management strategies, including demand regulation, infrastructure optimization, and improved resource allocation. The model can support decision-making by identifying high-risk periods and enabling scenario-based planning.

Finally, the integration of stress index forecasting with sectoral demand predictions provides a comprehensive framework for assessing system resilience under different conditions.

VI. Results and Discussion

A. Overall Model Performance

TABLE I
Prediction Performance on Validation and Test Sets

Variable	MAE	RMSE
Validation (2022–2025)		
Total Demand	1.12	1.44
Supply	1.92	2.27
Storage	3.96	4.50
Test (2026–2035)		
Total Demand	1.56	1.88
Supply	2.47	2.93
Storage	4.21	5.36

The model demonstrates strong predictive performance across all variables, with consistent results between validation and test datasets. This indicates good generalization and stability of the learned temporal patterns.

Demand is predicted with the highest accuracy, followed by supply, while storage remains the most challenging variable due to its cumulative and dependent nature.

B. Demand and Supply Forecasting

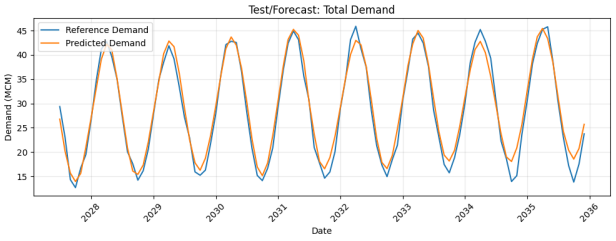


Fig. 2. Test set demand forecasting. The model accurately captures seasonal patterns and peak consumption behavior.

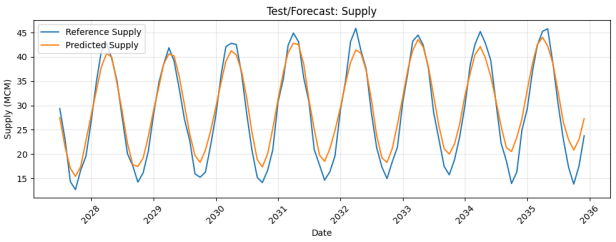


Fig. 3. Test set supply forecasting. The model reproduces seasonal variability with slight smoothing of extreme fluctuations.

The model successfully captures the strong seasonal structure of both demand and supply. Predicted curves closely follow observed values, with accurate reproduction of both peaks and troughs.

Supply prediction exhibits slightly higher error due to its dependence on climatic variability, particularly rainfall and temperature. However, the model preserves the overall temporal dynamics, making it suitable for long-term forecasting and scenario analysis.

C. Reservoir Storage Dynamics

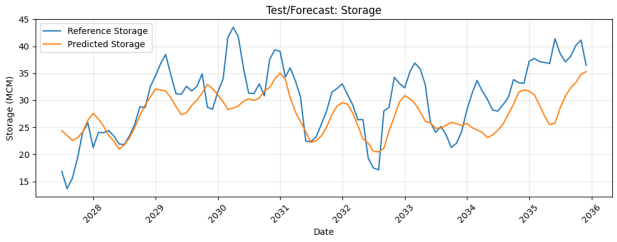


Fig. 4. Reconstructed storage based on predicted demand and supply. The predicted series is smoother and physically consistent.

Reservoir storage is modeled as a reconstructed variable derived from predicted demand and supply. The model captures the overall trend and seasonal evolution of storage, while maintaining physically realistic bounds.

However, predicted storage remains smoother than observed values. This smoothing effect reflects the model's tendency to minimize average error and highlights the inherent difficulty of modeling cumulative hydrological variables.

D. Sectoral Demand Analysis

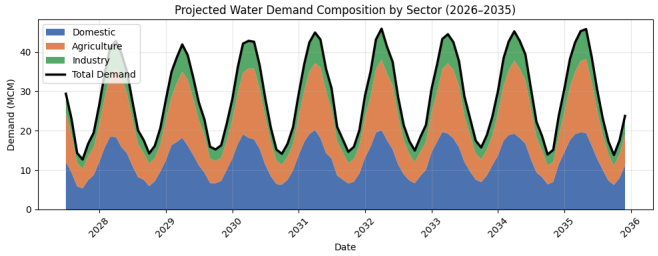


Fig. 5. Projected water demand composition by sector (2026–2035)

A decomposition of demand into domestic, agricultural, and industrial sectors provides deeper insight into consumption patterns.

Agricultural demand dominates total water consumption, followed by domestic demand, while industrial demand remains relatively stable. This structure remains consistent across the forecasting horizon.

1) Sectoral Forecasting Results: To improve interpretability, each sector is analyzed individually in the test period.

TABLE II
Sectoral Demand Prediction Performance

Sector	MAE	RMSE
Validation (2022–2025)		
Domestic	0.54	0.71
Agricultural	0.62	0.77
Industrial	0.24	0.31
Test (2026–2035)		
Domestic	0.73	0.93
Agricultural	0.87	1.02
Industrial	0.33	0.41

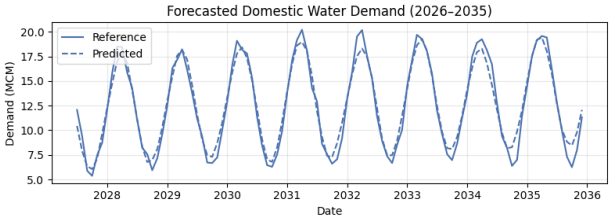


Fig. 6. Forecasted domestic water demand (2026–2035)

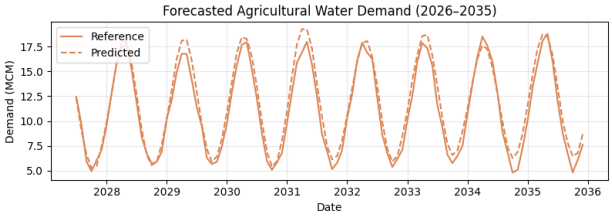


Fig. 7. Forecasted agricultural water demand (2026–2035)

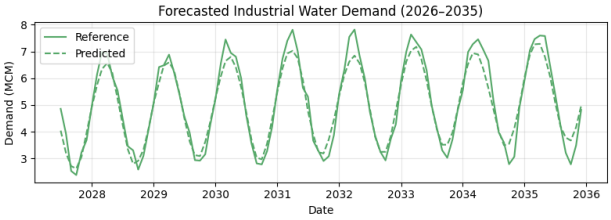


Fig. 8. Forecasted industrial water demand (2026–2035)

The results show that the model accurately captures heterogeneous sectoral behaviors:

- **Domestic demand** is predicted with high accuracy, reflecting stable population-driven consumption patterns.
- **Agricultural demand** exhibits strong seasonal variability, which is well captured despite slightly higher fluctuations.
- **Industrial demand** remains the most stable component and is predicted with very low error.

Overall, the model demonstrates the ability to learn distinct temporal dynamics across sectors.

E. Model Behavior and Limitations

The model shows strong generalization capabilities, supported by consistent performance across validation and test datasets. Regularization techniques such as dropout and early stopping contribute to stable training and prevent overfitting.

However, several limitations should be considered.

First, the model exhibits a smoothing effect, particularly in storage predictions, which may underestimate extreme variations. This is a common limitation of neural networks trained with regression-based loss functions such as Huber loss.

Second, storage prediction is inherently challenging due to its cumulative dependency on both demand and supply. Small prediction errors can accumulate over time, leading to deviation from actual trajectories.

Third, the dataset used is partially synthetic and includes reconstructed storage values. While this enables controlled experimentation, it may not fully reflect real-world variability.

Fourth, the model does not explicitly enforce physical constraints such as mass conservation. Incorporating physics-informed mechanisms could improve consistency and reliability.

Finally, the model produces deterministic predictions and does not quantify uncertainty. Future work should explore probabilistic approaches for risk-aware forecasting.

F. Practical Implications

The proposed framework provides a scalable and interpretable solution for water resource forecasting and planning.

By combining multivariate prediction, sectoral decomposition, and stress analysis, the model enables:

- Identification of high-demand periods
- Detection of potential water stress conditions
- Support for scenario-based planning

This makes the approach particularly valuable for decision-makers seeking to manage water resources under increasing climatic and demand pressures.

VII. Model Predictive Control for Water Allocation

While the LSTM model provides accurate forecasts of water demand, supply, and storage, it does not directly address how water should be allocated among competing sectors under limited resources. To bridge this gap, a Model Predictive Control (MPC) strategy is introduced as a decision-making layer on top of the forecasting model.

A. Problem Formulation

The objective of the MPC is to allocate available water resources among three sectors—domestic, agricultural, and industrial—while respecting system constraints and priority policies.

At each time step, the controller solves an optimization problem over a finite prediction horizon using forecasted values from the LSTM model. The objective is to maximize weighted allocations:

$$\max \sum_{i=1}^{H} (w_d x_{d,i} + w_a x_{a,i} + w_i x_{i,i}) \tag{2}$$

where $x_{d,i}$, $x_{a,i}$, and $x_{i,i}$ denote allocations for domestic, agricultural, and industrial sectors at time i, and w_d, w_a, w_i are the corresponding priority weights.

B. Constraints

The optimization is subject to the following constraints:

- **Water balance:**

$$s_{i+1} = s_i + \text{Supply}_i - \sum_j x_{j,i} - \text{Evaporation}_i \tag{3}$$

- **Storage constraint:**

$$s_i \geq S_{\min} \tag{4}$$

- **Allocation bounds:**

$$0 \leq x_{j,i} \leq \hat{D}_{j,i} \tag{5}$$

where $\hat{D}_{j,i}$ represents the forecasted demand for sector j.

C. Closed-Loop Evaluation

The MPC operates in a receding-horizon manner. At each step:

- The optimization is solved using forecasted demand and supply.
- Only the first allocation is applied.
- The system state (storage) is updated using actual observed values.

This closed-loop evaluation accounts for forecasting errors and provides a realistic assessment of system performance.

D. Allocation Results

The results of the MPC allocation over the test period (2026–2035) are summarized in Table III.

TABLE III
MPC ALLOCATION PERFORMANCE

Sector	Total Deficit (MCM)	Stress Months
Domestic	30.77	41
Agricultural	4.44	14
Industrial	20.78	59

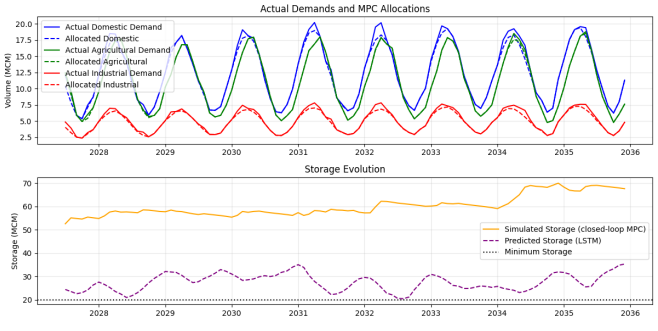


Fig. 9. Closed-loop MPC allocation results (2026–2035). Top: actual sectoral demands vs allocated volumes. Bottom: simulated storage under MPC compared to LSTM-predicted storage and minimum storage constraint.

E. Discussion

The results demonstrate that the MPC strategy effectively balances water allocation under constrained conditions.

Agriculture exhibits the lowest deficit and number of stress months, indicating efficient allocation under moderate priority. The industrial sector, assigned the lowest priority weight, experiences the highest number of stress months, reflecting its role as a flexible demand component.

Despite having the highest priority, the domestic sector still experiences some deficit. This behavior is attributed to forecasting errors, particularly when actual demand exceeds predicted values, limiting the controller's ability to allocate sufficient water.

Overall, the MPC framework ensures high satisfaction levels across all sectors while maintaining storage above the safety threshold. The results highlight the importance of combining accurate forecasting with optimization-based allocation to achieve realistic and effective water resource management.

F. Implications

The integration of MPC transforms the forecasting model into a complete decision-support system. It enables:

- Priority-based allocation of water resources
- Adaptation to demand and supply variability
- Robust performance under forecasting uncertainty

This approach provides a practical tool for operational water management and policy implementation.

VIII. Policy Scenario Analysis

To evaluate the behavior of the proposed model under different policy assumptions, three scenarios are considered: baseline, virtual water, and food self-sufficiency. Each scenario represents a distinct water management strategy and leads to different demand and supply dynamics.

The baseline scenario corresponds to the continuation of current conditions without major policy changes. As shown in Fig. 10, this scenario produces the highest demand levels across the forecasting period, with pronounced seasonal peaks. This behavior is consistent with steady population growth and unchanged consumption patterns. The baseline scenario exhibits the lowest prediction error, which can be attributed to its strong similarity with the historical data used for model training. Specifically, the demand MAE is approximately 1.37 MCM and the RMSE is 1.66 MCM, indicating that the model effectively captures the underlying temporal dynamics when future conditions remain close to those observed in the past.

The virtual water scenario assumes the importation of water-intensive goods, a strategy widely discussed as a means of reducing local water stress [8]. The results show a consistent reduction in demand levels compared to the baseline, while preserving the same seasonal structure. The model captures this shift with moderate degradation in accuracy (MAE $\approx$ 1.79 MCM, RMSE $\approx$ 2.18 MCM), suggesting that the changes introduced by this policy remain relatively smooth and predictable.

In contrast, the food self-sufficiency scenario leads to increased demand levels and greater variability. By promoting local agricultural production, this policy amplifies water consumption and introduces more irregular demand patterns. This observation is consistent with previous studies on water footprint and agricultural water use [9]. The increased complexity of the demand signal is reflected in a significant rise in prediction error (MAE $\approx$ 3.65 MCM, RMSE $\approx$ 4.55 MCM),

indicating that these patterns deviate more strongly from the historical data used for training.

Across all scenarios, supply prediction errors exhibit smaller variations compared to demand. This reflects the fact that supply is primarily driven by climatic variables, whereas demand is more sensitive to policy and socio-economic factors. Consequently, policy interventions have a more pronounced impact on demand than on supply.

TABLE IV
PREDICTION PERFORMANCE UNDER DIFFERENT POLICY SCENARIOS

Scenario	D-MAE	D-RMSE	S-MAE	S-RMSE
Baseline	1.37	1.66	2.01	2.60
Virtual Water	1.79	2.18	2.56	3.10
Food Self-Sufficiency	3.65	4.55	2.45	3.02

Table IV summarizes the prediction performance across the three scenarios. The baseline scenario achieves the lowest error due to its similarity to historical patterns, while the food self-sufficiency scenario shows the highest demand prediction error, reflecting increased variability and system stress.

Reservoir storage, reconstructed from demand and supply, is indirectly influenced by these changes. In scenarios with increased demand variability, such as food self-sufficiency, larger fluctuations in storage are expected, which may amplify the risk of water stress during peak periods.

Figure 10 illustrates the predicted demand trajectories under the three scenarios. While seasonal patterns are preserved, clear differences in demand magnitude emerge depending on the policy context.

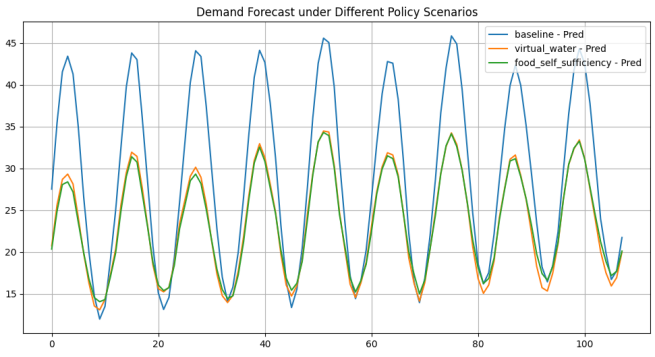


Fig. 10. Predicted water demand under baseline, virtual water, and food self-sufficiency scenarios. Seasonal patterns are preserved across scenarios, while demand levels vary depending on the policy context.

Beyond predictive accuracy, it is essential to assess the impact of each policy on overall system performance. The virtual water scenario reduces total demand, leading to lower stress index values and fewer deficit periods. This indicates an improvement in water security, as the system operates further from critical thresholds.

Conversely, the food self-sufficiency scenario increases demand variability and amplifies deficit conditions. Higher demand levels result in more frequent exceedance of the critical stress threshold, reflecting increased pressure on available resources and reduced system resilience.

Overall, these results demonstrate that policy choices have a direct and significant impact on water system sustainability, extending beyond their influence on model prediction accuracy.

IX. FRAMEWORK FOR WATER RESOURCE MANAGEMENT (MEWRE)

Based on the results of the proposed deep learning framework, including multivariate LSTM forecasting, stress analysis, and policy scenario evaluation, a data-driven decision-support system is proposed for the Ministry of Electricity, Water and Renewable Energy (MEWRE).

A. AI-Driven Demand Management

The deep learning model identifies water demand, particularly in the agricultural sector, as the primary driver of system stress. Leveraging these predictive insights, the following strategies are recommended:

- Use model forecasts to anticipate high-demand periods and implement preemptive demand reduction strategies.
- Promote water-efficient irrigation techniques guided by predicted demand peaks.
- Encourage virtual water strategies to reduce local demand for water-intensive goods.

This enables proactive demand management based on data-driven predictions.

B. Supply Optimization Using Predictive Insights

The model provides accurate forecasts of supply dynamics based on climatic variables, which can be used to optimize resource allocation.

- Adjust supply planning based on predicted seasonal variability.
- Reduce losses in distribution systems to increase effective supply.
- Integrate desalination and reuse strategies aligned with forecasted deficits.

These actions improve system efficiency using predictive intelligence.

C. Priority-Based Allocation Strategy

A priority-based allocation system, informed by model predictions, is recommended to manage scarcity conditions.

- Domestic demand is prioritized to ensure essential needs.
- Agricultural demand is dynamically adjusted based on predicted availability.
- Industrial demand is reduced during high-stress periods identified by the model.

This strategy translates model outputs into actionable allocation policies.

D. Real-Time Monitoring and Early Warning System

The stress index derived from model predictions can be used as an early warning mechanism.

- Monitor predicted stress levels to identify upcoming critical periods.
- Trigger early interventions when stress exceeds predefined thresholds.
- Support adaptive decision-making based on continuous model updates.

This shifts the system from reactive to predictive management.

E. Policy Evaluation Using Scenario Simulation

The deep learning framework enables the simulation of different policy scenarios and their impact on system stability.

- Virtual water strategies reduce demand and improve system resilience.
- Food self-sufficiency strategies increase demand variability and system stress.

These simulations allow policymakers to evaluate trade-offs before implementation.

F. Long-Term Resilience Planning

The model supports long-term planning by providing reliable forecasts under different conditions.

- Use long-term predictions to guide infrastructure investments.
- Integrate climate projections into the forecasting pipeline.
- Combine deep learning with physical models for improved reliability.

This ensures robust planning under uncertainty.

G. Summary

The proposed framework transforms the deep learning model into a practical decision-support tool for MEWRE. By integrating predictive analytics, sectoral insights, and policy evaluation, it enables data-driven, adaptive, and sustainable water resource management.

X. Conclusion

This work presented a comprehensive framework for water resource forecasting and management by integrating multivariate deep learning, optimization, and policy scenario analysis. A Long Short-Term Memory (LSTM) model was developed to jointly predict water demand, supply, and reservoir storage, effectively capturing nonlinear temporal dependencies and seasonal patterns across multiple sectors. The results demonstrated strong predictive performance on both validation and test periods, with demand variables achieving higher accuracy than supply and storage due to their more stable underlying dynamics.

Beyond forecasting, a Model Predictive Control (MPC) strategy was introduced to transform predictions into operational decisions. The proposed control framework dynamically allocates water across domestic, agricultural, and industrial sectors while maintaining storage above safety thresholds. Experimental results showed that the MPC approach effectively balances competing demands, reduces deficit conditions, and ensures system stability under varying scenarios.

The analysis of policy scenarios further highlighted the critical role of decision-making in shaping system behavior. While the baseline scenario yielded the lowest prediction error, alternative strategies such as virtual water trade and food self-sufficiency significantly altered demand patterns and system stress levels. In particular, increased demand variability under self-sufficiency policies led to higher prediction errors and greater pressure on available resources, emphasizing the importance of adaptive and robust management strategies.

Overall, this study demonstrates that sustainable water resource management requires an integrated approach combining accurate forecasting, optimization-based control, and policy-aware analysis. The proposed framework provides a practical and interpretable decision-support tool for anticipating future conditions, improving resource allocation, and mitigating water stress.

Future work may extend this framework by incorporating additional climatic and socio-economic variables, improving uncertainty quantification, and exploring reinforcement learning-based control strategies for enhanced adaptability in highly dynamic environments. Such an integrated perspective is essential for designing resilient water systems capable of adapting to increasing climatic variability and growing resource constraints.

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